YASIR RAHMAN

+8801323934498 Email LinkedIn GitHub Google Scholar Portfolio

About Me

I'm a **Software Engineer** and **Researcher** with interests in **security vulnerability analysis** and **trustworthy systems**, focused on analyzing vulnerabilities in complex systems and developing solutions to build **reliable and trustworthy systems**. I aim to pursue a **PhD** in this area to bridge research with practical impact.

Education

Chittagong University of Engineering and Technology *Feb, 2020 - Jun, 2025*
BSc in Computer Science and Engineering *CGPA: 3.53 / 4.00*

- Awarded **merit scholarships in all 8 semesters** and represented CUET in **ICPC Dhaka Regional Contests**.

Test Scores

GRE: Quantitative: **169**, Verbal: **155**, Analytical Writing: **4.0**
IELTS: **8.0** (Listening: **8.5**, Reading: **8.5**, Writing: **7.0**, Speaking: **7.5**)

Research Experience

LLMs as Validators for Cross-Chain Bridges: Evaluating Source-Level and Transaction-Level Attack Detection on Solana *July 2026 - Present*
Ongoing Research *In collaboration with Dr. Shahnewaz Karim Sakib and Dr. Anindya Bijoy Das*

- Designed and built a **cross-chain bridge on Solana in Rust**, planted four security flaws (two modelled on the Wormhole and Cashio hacks), and wrote working exploits for each, verified against a secure baseline.
- Designed and ran an **8,000-run experiment** testing whether AI models can detect these attacks: they found 57% of the planted bugs by reading code, but under 1% of three of the four attacks from transaction records.
- Developed a blind evaluation pipeline in which two independent AI graders (Claude, Gemini) scored 3,200 answers with 99.4% agreement, replacing an unreliable keyword-based grader.

Memory Injection Attacks on Long-Term Memory of LLM Agents *July 2026 - Present*
Ongoing Research *In collaboration with Dr. Shahnewaz Karim Sakib and Dr. Anindya Bijoy Das*

- Investigating existing **memory injection attacks** on LLM agents' long-term memory and their limitations.
- Analyzing LLM agent memory vulnerabilities to improve both attack strategies and defense mechanisms.

Attack Patterns in Cross-chain Bridges: A Trust-based Empirical Analysis *January 2026 - Present*
Manuscript Under Review *In collaboration with Dr. Rahma Bintey Mufiz Mukta*
GitHub Repository: https://github.com/ChyYasir/cross-chain-bridge-attack-patterns

- Conducted a comprehensive survey and taxonomy of security vulnerabilities and attack patterns in cross-chain communication protocols.
- Collected and analyzed **50+ real-world cross-chain bridge attack incidents** from 2022 to August 2026, drawing from sources such as bridges' own post-mortem reports and third-party security audits.
- Mapped attacks to specific attack patterns and functional domains based on analysis.

Predictive Model for Birth Delivery Mode Using EHG Signals *September 2023 - November 2025*
Research Assistant *In collaboration with Dr. Rubana Hoque Chowdhury*
GitHub Repository: https://github.com/ChyYasir/Predictive-Model-for-Birth-Delivery-Mode

- Designed and implemented **machine learning pipelines in Python** for predicting birth delivery mode from electrohysterogram (EHG) signals, covering signal preprocessing, feature extraction, model training, and evaluation.
- Re-implemented methods from prior published EHG studies as baselines and ran iterative experiments to refine models.

Bengali Keywords Extraction from Bengali Blog Using Transformer-Based Approach *2024 - 2025*
Undergraduate Thesis Research *Supervised by Dr. Muhammad Ibrahim Khan*

- Developed a comprehensive Bengali blog dataset of 9,012 annotated posts and proposed a transformer-based dual-encoder architecture achieving 84% F1@10 score and did some comparison analysis with the existing methods.

Publications

[2] (2026). *Attack Patterns in Cross-chain Bridges: A Trust-based Empirical Analysis.* Under review at **Blockchain: Research and Applications**. Preprint: arXiv:2611.01587

[1] (2026). *Analysis of electrohysterogram signals for predicting obstetric outcome using machine learning methods: a scoping review.* **BMC Pregnancy and Childbirth**, Springer Nature (**Q1**). doi.org/10.1186/s12884-026-09491-6

Teaching Experience

International Islamic University Chittagong (IIUC) *February 2026 - August 2026*
Adjunct Lecturer, Dept. of Computer Science and Engineering *Chittagong, Bangladesh*

- Taught undergraduate courses in **Algorithms** (theory & lab), **Compiler Design** (theory & lab), and **Competitive Programming II** (lab).
- Developed a content website, Yasir Explains, to supplement course material for students.

Industry Experience

Bevy Commerce
July 2024 - Present

Software Engineer (L3) *Ontario, Canada (Remote)*

- Contributed to the architecture and core workflows of a platform for the peptide industry (880K+ lines of code). A key contribution was reimplementing login from single-account to **multi-binding login**, impacting around **100,000 users**.
- Developed a **data migration tool** to move data from **Magento** and **NetSuite** to **Shopify** using **Shopify GraphQL mutations**, successfully migrating over **5 million data points**.
- Built **SFTP**-based file transfer pipelines and automation scripts for client data migrations, cutting down manual intervention and reducing failed transfers.

BreakByte
December 2023 - June 2024

Software Engineer *Chittagong, Bangladesh*

- Implemented secure authentication and authorization for a full-stack blogging platform and integrated the **Tiptap** editor for rich content creation.
- Developed a scalable **multi-tenant architecture**, reducing onboarding time by approximately **50%**, and deployed the platform on **AWS EC2/S3** for secure, scalable infrastructure.

Technical Skills

- **Languages:** Python, C, C++, JavaScript
- **ML Tools and Libraries:** PyTorch, Scikit-learn, Pandas, NumPy, TensorFlow
- **Frameworks:** React, Node.js, Express.js, Next.js
- **Databases:** MongoDB, PostgreSQL, MySQL, Redis
- **Cloud and Tools:** AWS (EC2, S3), Docker, Git

Projects

Rilo EMS (Employee Management System) *Next.js, TypeScript, MongoDB, ShadCN*

GitHub Repository: https://github.com/ChyYasir/rilo-ems *December 2024 - January 2025*

- **Led an 8-member development team** to design and ship a full-featured employee management system in **Next.js**, owning the architecture and code-review process that kept the codebase consistent across all 8 contributors.
- Implemented the **backend boilerplate** and **authentication system**, and integrated the core tools, libraries, and **MongoDB** database that the rest of the team built on.

Easy Tuition (Tuition Management System) *MERN Stack, Material UI*

GitHub Repository: https://github.com/ChyYasir/EasyTution

Live Demo: https://easy-tution.onrender.com/ *October 2023 - June 2024*

- Built a **MERN-stack** admin dashboard with a tutor-matching algorithm and optimized search/filtering across **10,000+ student and tutor records**, cutting administrative decision-making time from **3+ hours to 5-10 minutes**.

Awards & Achievements

- **ICPC Regionalist, Dhaka Region:** 2021-22 (Rank 48) & 2022-23 (Rank 41) | View Certificate
- **East Delta University Engineering Day Programming Contest 2022:** Rank 1
- **CUET Inter University Programming Contest 2022:** Rank 10
- **BUET Inter University Programming Contest 2022:** Rank 47
- **Sylhet Inter University Junior Contest:** Rank 12
- **2nd Runner-up:** CUET Intra Junior Programming Contest 2021 (Rank 3)
- **Top 16 Finalist:** BUET CSE FEST 2024 Hackathon (National) | GitHub
- **2nd Runner-up:** Electrical Surge 2.0 Hackathon (CUET IEEE)
- **2nd Round:** DEVOPS Hackathon, Building Platform Using Cloud (Poridhi, Powered by Brain Station 23)
- **Solved 2000+ problems** across online judges: Codeforces: CHY_YASIR, AtCoder: Chy_chy
- **Selected to Represent Bangladesh** at the International Olympiad on Astronomy and Astrophysics (IOAA) 2017

Co-curricular Activities

CUET Computer Club *2022 - 2025*

Instructor & DSA Mentor, CP Bootcamp / Competitive Programming Group *Chittagong, Bangladesh*

- Mentored students on Data Structures & Algorithms (Graphs, Segment Trees, Bitmask, etc.) for ICPC and university contests. **Recorded Sessions:** YouTube Channel
- Designed **competitive programming problems** and organized contests for ICPC and IUPC preparation, followed by post-contest discussions.

IEEE Computer Society, CUET Student Branch *2024 - 2025*

Vice Chair *Chittagong, Bangladesh*

- Organized an intra-CUET microservice hackathon and a competitive programming contest, and conducted multiple educational sessions on software engineering and competitive programming.